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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/575,507	12/29/2023	Lutz SCHOMBURG	GRUDAN00012USU1	3645

20210 7590 06/29/2026

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EXAMINER
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ART UNIT	PAPER NUMBER
1677	

NOTIFICATION DATE	DELIVERY MODE
06/29/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/575,507	Applicant(s) SCHOMBURG, Lutz	
	Examiner ELLEN J MARCSISIN	Art Unit 1677	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12/29/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 12/29/2026 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Priority

2. The present application was filed as a proper National Stage (371) entry of PCT Application No. PCT/EP2022/067808, filed 06/28/2022. Acknowledgment is also made of applicant's claim for foreign priority under 35 U.S.C. 119(a)-(d) to Application No. 102021206749.2, filed on 06/29/2021 and 102021209147.4, filed 08/19/2021, in Germany.

Information Disclosure Statement

3. The information disclosure statement (IDS) filed 12/10/2024 is considered, initialed and is attached hereto.

Claim Objections

4. Claims 2-5 and 12-20 are objected to because of the following informalities:

Claims 2-5 and 12-20 recite the language "method according to claim…", it is suggested that Applicant amend the claims to recite language such as "The method of claim…", in order to improve clarity of the record, the suggested language clearly referring back to the claim from which the claims depend from, and clearly indicating that all the previous limitations are encompassed/included.

Appropriate correction is required.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claims 1-10 and 12-20 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 1 recites at the preamble the method "for evaluating the health risk of selenium (Se) deficiency in a subject", the claim language is indefinite because it is unclear from the recited language if the claimed assay steps evaluate the risk of potential selenium deficiency (predicting one's risk of Se deficiency, as in future risk), or rather a health risk which is unspecified that is a result of a subject's selenium deficiency (i.e., a risk of some other health status, as in condition, as a result of selenium deficiency). Further it is not clear what is and is not encompassed by the language "evaluating a health risk", for example the boundaries are unclear

because it is not readily understood what is and is not encompassed by "evaluating". Specifically evaluating could merely encompass merely observing the presence of the targeted analytes in a subject who is deficient, or contrary to this interpretation could mean performing some type of diagnostic/prognostic determination. As a result, it is not clear if this language imparts any particular or specific action to be taken, other than merely performing the steps recited at claim 1 following the preamble, particularly considering the claim merely recites the act of determining measured values for the indicated biomarkers, and recites no conclusory steps/actions to be taken that relate back to the preamble in a way that clarify what is encompassed by the preamble.

Similarly as above, claims 3 and 12 recite "correlated with the risk related to Selenium deficiency", it is unclear what "risk" this is in reference to, as the claims do not clearly specify what "health risk" related to Selenium the claims are evaluating (risk of what condition).

Claim 2 recites the limitation "setting two or more biological parameters into correlation" and "wherein said biological parameters are represented by the said measured biological values obtainable as defined in steps a) and b). The recited language is indefinite because it is not understood what is meant by "setting…into correlation", i.e., there is no indication as to what action is or is not "setting into correlation", as the word "setting" suggests this is an action taken by the practitioner.

Furthermore, "wherein said biological parameters are represented by said measured values obtainable as defined in steps a) and b)", is indefinite claim language because it is not clear if this language is intended to limit the parameters to those determined biological values as recited at claim 1, or something similar, yet distinct because the claim uses language such as "represented by" and "as defined". The language "using the values obtained as defined in step a) and b) of claim 1 or ii) in step c) further suggests that applicant recites the values of a) and b)

(determined values) as distinct from c), (c being parameters "obtainable as defined" in steps a) and b). This language raises question as to what the similarity or difference between the "determined values" and the "set parameters".

Claim 2 also recites "calculating a risk index score…using the values obtainable as defined i) in step a) and b) of claim 1 or ii) in step c", however, there is no indication in the claims as to what this calculation involves, or how the "values" are used (no indication as to what equation or algorithm is used to "calculate a risk index score"). The recited language is indefinite because it suggests some calculation or formula without indicating what that calculation or formula is. As a result, the claim is indefinite because it is not clear what would and would not be encompassed by "calculating a risk index score".

Claim 2 recites the limitation "the values obtainable as defined… or ii) in step c" in step d. Specifically, only steps a) and b) refer to values ("determining…a measured biological value", and step c of claim 2 refers to set parameters, i.e., biological parameters represented by "said measured biological values". The claim language is confusing because it is unclear if step 3 is intended as calculating risk score using values of a) and b) which are the determined measured biological values, or if the values of step c) are also the determined biological values, or rather are the set parameters represented by the biological values.

Claims 3 and 12 recite "wherein said subject is suffering from a disorder selected from…." The claim listing a large list of possible disorders. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claims 3 and 12 recite the broad recitation "infection" "inflammation", "cancer", "severe illness treated

in intensive care unit (ICU)", "a neurodegenerative", and the claims also recites specific infections such as "sepsis", "SIRS", conditions associated with inflammation such as "inflammatory bowel disease", "sepsis", "SIRS", etc., specific cancers such as "breast", "prostate", colorectal", "lung", "liver", specific neurodegenerative diseases including "Alzheimer's", "Parkinson's, "Huntington's", etc., each of which are the narrower statement of the broader limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

Claim 6 is indefinite because the claim recites at the preamble, "The method of using the method as defined in claim 1". There is insufficient antecedent basis for the limitation "**The method** of using the method" in line 1. The claim is presented as a dependent claim, however, appears to be reciting the method for each of these claims as a new/separate method invention other than the method of claim 1. It is suggested that applicant amend the language in order to recite something such as "The method of claim 1, wherein the method is applied as a companion diagnostic method to monitor Se supplementation in a subject", and further recite a specific active method step directed to this recited intended use (i.e., a method step that amounts to applying the method for the claimed act of monitoring), or clearly recite the methods as separate independent claims.

Claims 7-10 are similarly recited as being indefinite for reciting "The method of using the method as defined in claim 1", for the reasons as indicated above (referring to claim 6), as the same reasoning also applies presently.

Additionally, regarding claims 7-10, each of these claims recites "The method of using the methods" to perform an additional intended use, however none of these claims set forth any

additional steps involved in the additional "uses". See MPEP 2173.05, attempts to claim a process without setting forth any steps involved in the process generally raises indefiniteness issue. In the present case, there are no steps set forth that distinguish the methods of these claims from the method of claim 1, which is for evaluating health risk of selenium deficiency. Therefore, it is not readily clear if additional steps/actions are required by the claimed intended use, beyond the assay steps as performed at claim 1 (i.e., it is not clear if the "use" limitation implies or suggests additional active actions to be taken as part of the process). Nonetheless, these claims are given broadest reasonable interpretation and are addressed below under 35 U.S.C. 103, as well.

Claims 15-19 recites the limitation "The method of using the risk index score obtainable according to claim 2" in lines 1-2. There is insufficient antecedent basis for this limitation in the claim. Specifically, the claim present no earlier claim directed to "a method of using the risk score" as recited at claim 15. Additionally, the language "using the risk score obtainable to claim 2" is language that also does not clearly require the act of obtaining a risk score as recited at claim 2.

Claim 9 recites "sepsis, infection…SIRS", the recited language is indefinite because "infection" is a broader term that appears to encompass each of "sepsis" and "SIRS" (each sepsis and SIRS are themselves considered specific types of infections).

Claim 10 recites "suffering from a disease selected from viral infection including influenza A and COVID-19", it is unclear if the claims are limited to viral infection that is one of influenza A or COVID-19, or rather if these specific species of viral infection are exemplary. See MPEP 2173.05(d), because the scope of what is encompassed by the recited language is unclear, the claims is indefinite.

Claim 16 recites the limitation "the individual Se requirement and dosage regime during therapeutic or prophylactic Se supplementation in a subject" in lines 1-2. There is insufficient antecedent basis for this limitation in the claim. The claims fail to earlier recite a limitation directed to “an individual Se requirement and dosage regime during therapeutic or prophylactic Se supplementation in a subject”. As such, it is not readily clear what the claim is in reference to by “the individual Se requirement and dosage regime during therapeutic or prophylactic Se supplementation in a subject”.

The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

> Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

6. Claim 9 is rejected on the basis that it contains an improper Markush grouping of alternatives. See *In re Harnisch*, 631 F.2d 716, 721-22 (CCPA 1980) and *Ex parte Hozumi*, 3 USPQ2d 1059, 1060 (Bd. Pat. App. & Int. 1984). A Markush grouping is proper if the alternatives defined by the Markush group (i.e., alternatives from which a selection is to be made in the context of a combination or process, or alternative chemical compounds as a whole) share a “single structural similarity” and a common use. A Markush grouping meets these requirements in two situations. First, a Markush grouping is proper if the alternatives are all members of the same recognized physical or chemical class or the same art-recognized class, and are disclosed in the specification or known in the art to be functionally equivalent and have a

common use. Second, where a Markush grouping describes alternative chemical compounds, whether by words or chemical formulas, and the alternatives do not belong to a recognized class as set forth above, the members of the Markush grouping may be considered to share a "single structural similarity" and common use where the alternatives share both a substantial structural feature and a common use that flows from the substantial structural feature. See MPEP § 2117.

The Markush grouping of sepsis, infection, (poly)trauma or SIRS is improper because the alternatives defined by the Markush grouping do not share both a single structural similarity and a common use for the following reasons: for example, although sepsis and SIRS are considered infections, poly(trauma) is a term that is broader in scope and is not limited to damage/injury that is infection.

To overcome this rejection, Applicant may set forth each alternative (or grouping of patentably indistinct alternatives) within an improper Markush grouping in a series of independent or dependent claims and/or present convincing arguments that the group members recited in the alternative within a single claim in fact share a single structural similarity as well as a common use.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

7. Claims 1-10 and 12-20 are rejected under 35 U.S.C. 101 because the claimed invention is directed to laws of nature/natural phenomenon and abstract ideas without significantly more.

The U.S. Patent and Trademark Office recently revised the MPEP with regard to § 101 (see the MPEP at 2106). Regarding the MPEP at 2106, in determining what concept the claim is "directed to," we first look to whether the claim recites:

(1) any judicial exceptions, including certain groupings of abstract ideas (*i.e.*, mathematical concepts, certain methods of organizing human activity such as a fundamental economic practice, or mental processes); and

(2) additional elements that integrate the judicial exception into a practical application (see MPEP § 2106.05(a)-(c), (e)-(h)).

Only if a claim (1) recites a judicial exception and (2) does not integrate that exception into a practical application, do we then look to whether the claim contains an "'inventive concept' sufficient to 'transform'" the claimed judicial exception into a patent-eligible application of the judicial exception. *Alice*, 573 U.S. at 221 (quoting *Mayo*, 566 U.S. at 82). In so doing, we thus consider whether the claim:

(3) adds a specific limitation beyond the judicial exception that is not "well-understood, routine, conventional" in the field (see MPEP § 2106.05(d)); or

(4) simply appends well-understood, routine, conventional activities previously known to the industry, specified at a high level of generality, to the judicial exception.

See MPEP 2106.

ELIGIBILITY STEP 2A: WHETHER A CLAIM IS DIRECTED TO A JUDICIAL EXCEPTION

Step 2A, Prong 1

Claim 1 recites "A method for evaluating the health risk of selenium (Se) deficiency in a subject", the method comprising "determining… a measured value….selenium (Se) and/or… selenoprotein P (SELENOP); and …antibodies against SELENOP (SELENOP-Ab)".

The natural relationship to which the claims are directed (i.e., the relation between Se, SELENOP, and SELENOP-Ab and Se deficiency) is a law of nature. Similar concepts have been held by the courts to constitute law of nature/natural phenomenon, as in the identification of a correlation between the presence of in a bodily sample (such as blood or plasma) and cardiovascular disease risk in *Cleveland Clinic Foundation v. True Health Diagnostics*, LLC, 859 F.3d 1352, 1361, 123 USPQ2d 1081, 1087 (Fed. Cir. 2017). In *Mayo,* the Supreme Court found that a claim was directed to a natural law, where the claim required administering a drug and determining the levels of a metabolite following administration, where the level of metabolite was indicative of a need to increase or decrease the dosage of the drug. See *Mayo Collaborative Services v. Prometheus Labs., Inc.*, 566 U.S. 66, 74 (2012).

The instant claims are similar in *Mayo* as they involve a "relation itself [which] exists in principle apart from any human action" (id. at 77), namely the relationship between the naturally occurring levels of each of Se, SELENOP and SELENOP-Ab and the presence of Se deficiency (health risk of Se deficiency). The correlation is a judicial exception as it exists in principle apart from any human action; the correlation itself therefore cannot form the basis for eligibility.

Claims 6-10 further recite "The method of using the method as defined in claim 1…", the claims reciting specific further intended uses, namely "as a companion diagnostic method to monitor Se supplementation in a subject" (claim 6), monitoring Se supplementation further, when given broadest reasonable interpretation, encompasses mental processes, i.e., concepts performed in the human mind, such as after obtaining the assay results observing, evaluating or

performing a judgment or opinion regarding supplementation of Se. Claim 7 recites "using… in the quantitation of the individual Se requirement and dosage regime during therapeutic or prophylactic Se supplementation in a subject ", which also encompasses mental processes, forming an observation, evaluation, judgment or opinion, as there are no specific active method steps further recited, required to be performed as part of the method. This is further the case for claims 8-10, "using…in the analysis and monitoring of Se poisoning or the treatment of a subject suffering from selenosis by administration of a pharmaceutically effective amount of an antibody against SELENOP" (claim 8), using.. in the treatment of a subject suffering from a disease selected from sepsis, infection, poly(trauma), or SIRS" (claim 9), "using…in the treatment of a subject suffering from a disease selected from viral infection including influenza A and COVID-19" (claim 10). Given broadest reasonable interpretation, none of these limitations go beyond mental concepts, for example such a subject performing the assay and thinking about the results, as indicated above.

Claim 2 recites, "further comprising the steps of: c) setting two or more biological parameters into correlation, wherein said biological parameters are represented by the said measured biological values obtainable as defined in steps a) and b) of claim 1; and d) calculating a risk index score for the Se deficiency of said subject using the values obtainable as defined i) in step a) and b) of claim 1 or ii) in step c)". The claim recites an abstract idea, specifically the act of "setting into correlation", "calculating a risk index score" is considered to be an abstract idea as these limitations encompass mathematical concepts, namely mathematical relationships, calculations (see Groupings of Abstract Ideas, 2106.05).

Claims 15-19 are substantially similar to claims 6-10 discussed above, the claims differing in that they recite "using the risk index score obtainable in claim 2…" in the various

intended uses (i.e,. the claim encompass both mathematical concepts, referring to the calculation at claim 2, and the mental concepts indicated above). The analysis is substantially similar to that above, the claims amount to no more than concepts performed in the human mind (abstract ideas are mental concepts).

Step 2A, Prong 2

The above discussed limitations are themselves the judicial exceptions, and as such, are not sufficient to amount to integration into a practical application of the judicial exception. Rather "integration into a practical application" requires an additional element or combination of elements in the claim to apply, rely on or use the judicial exception in a manner that impose meaningful limit on the judicial exception.

In the present claims, regarding claim 1, beyond the correlation itself, the claim recites merely "determining" measured biological values for Se and/or SELENOP, and SELENOP-Ab. These limitations merely amount to obtaining the data (mere data gathering step necessarily performed). The determining steps are recited at an extremely high level of generality, and are not, for example limited to any particular unconventional testing technique/reagent or platform, as claimed.

Further limiting claim, claims 4 and 5 recite types of assays, however, these claims are still extremely general, and importantly still directed to steps of gathering the data. These limitations do not go beyond insignificant presolution activity, i.e., a mere data gathering step necessary to use the correlation, similar to the fact pattern in *In re Grams*, 888 F.2d 835 (Fed. Cir. 1989) and *Ariosa Diagnostics, Inc. v. Sequenom, Inc*. (Fed. Cir. 2015).

Claim 3 recites "wherein the said measured biological values are correlated with risk related to Selenium deficiency in said subject, wherein said subject is suffering from a disorder

selected from…", the claim lists many disorders. As presented, the claims are limited to performing a method correlating the results with deficiency in a subject population as limited by claim 3 (not correlating with any of those disorders at claim 3). The limitations further limit the subject population, which narrows the judicial exception, however, does not amount to an integration of the judicial exception into a practical application thereof.

This is similarly the case regarding claim 12, which further limits the subject of claim 2.

None of the claims recited additional limitations, in addition to the judicial exceptions themselves which apply, rely on or use the judicial exceptions in a meaningful way, so to overcome the rejection.

Although claims 7-9 and 16-18 recite limitations that reference dosage regimes and treatment by administration of a therapeutic (i.e., an antibody against SELENOP), none of these claims recite steps directed to administering a treatment as a result of the judicial exceptions (i.e., fail to apply a particular and specific treatment as a clear result of the judicial exception in a way sufficient to amount to integration of the judicial exceptions into a practical application thereof).

ELIGIBILITY STEP 2B: WHETHER THE ADDITIONAL ELEMENTS CONTRIBUTE AN "INVENTIVE CONCEPT"

Further, the additional limitations discussed above (claims 3 and 12), 4 and 5, fail to amount to more than that which well-known routine and conventional in clinical/assay art. The claims, for example, fail to recite any particular testing technique/reagent, etc. which goes beyond that which well well-known routine and conventional. The claims, as noted above, with respect to the "determining…value" steps are extremely general. Such steps do not go beyond insignificant extra-solution activity in that these are mere data gathering steps necessary in order to use the judicial exceptions.

Further regarding the routine and conventional nature of the "determining steps", see for example, pages 4-5 of the originally filed specification, which acknowledge that detection methods/methods of determining the claimed biomarkers, available and known to those of ordinary skill in the prior art.

For all these reasons, the claimed invention is rejected under 35 U.S.C. 101.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

8. Claim(s) 1 and 4-7 are rejected under 35 U.S.C. 103 as being unpatentable over Sun et al., "Autoimmunity to Selenoprotein P in thyroid patients", Eur. Thyroid J. 42nd Annual Meeting of the ETA, 2019 (IDS entered 12/10/2024) in view of Ventura et al., Selenium and Thyroid Disease: From pathophysiology to Treatment, International Journal of Endocrinology, Vol. 217 (Article ID 1297658), (2016), (9 pages).

Sun et al. suggest autoimmunity (autoantibodies) to SELENOP as contributing to the development of ATTD (autoimmune thyroid disease) (paragraph 1). Sun teach performing a qualitative assay for serum aAB to SELENOP, measured in thyroid patients and healthy controls, assessing Se concentration by total reflection X-ray fluorescence (TXRF), serum SELENOP by ELISA (see paragraph 2). Sun teach prevalence of SELENOP-aAB in thyroid patients higher than in controls, that SELENOP-aAB positive samples precipitated measurable Se concentration,

in contrast to control immunoglobulins. Sun teaching their results indicate that SELENOP-aAB affect Se and SELENOP, and may be relevant in AITD.

Sun's methods cited above teach determining in a sample (serum) of a subject a measured biological value for selenium, SELENOP and antibodies against SELENOP.

Sun et al. differs from the claimed invention in that Sun fails to teach evaluating the health risk of selenium (Se) deficiency in a subject (interpreted as evaluating risk of potential/subject's status regarding Se deficiency).

Ventura et al. teach that in the thyroid, selenium is required for antioxidant function and for the metabolism of thyroid hormones (abstract). Ventura also acknowledge that it is known int eh prior art that selenium intake has been associated with autoimmune disorders, literature suggesting that selenium supplementation of patients with AITD is associated with reduction in antithyroperoxidase antibody levels, improved thyroid ultrasound features, and improved quality of life (abstract), that supplementation in Grave's is associated with improvement of quality of life and eye involvement, as well as delayed progression of ocular disorders.

Ventura et al. teach selenium deficiency decreases synthesis of thyroid hormones, as it decreases the function of selenoproteins (page 3, end of col. 1 to col. 2). Ventura also suggest that studies seem to demonstrate that selenium deficiency is associated with higher prevalence of thyroid disease (page 6, col. 1, para 4). Ventura teach supplementation should be recommended to patients with low levels of selenium, that on the other hand, high selenium intake in individuals without proved deficit may have important adverse effects such as hyperglycaemia and atherosclerosis (page 7, col. 1, para 3).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the method of Sun et al.,

(detecting values for Se, SELENOP and aAB to SELENOP) in order to apply the method in order to evaluate the health risk of selenium (Se) deficiency in a subject, one motivated to do so because the prior art recognized that those with ATTD benefit from Se supplementation, however that there is a range in which Se levels are beneficial (supplementation should be recommended to patients with low levels of selenium, that on the other hand, high selenium intake in individuals without proved deficit may have important adverse effects to one's health, such as hyperglycaemia and atherosclerosis, see Ventura). As a result, one having ordinary skill in the art would be motivated to look to supplementation as a treatment for AITD (as a result of the recognized advantages observed in those with ATDD receiving supplementation), and would be motivated to monitor levels to achieve supplementation where deficient as needed, but without risk of additional adverse effects (evaluating health risk, consistent with the present claims).

One having ordinary skill in the art would have a reasonable expectation of success because each of the assays for determining/measuring the levels of Se, SELENOP, and aAB-SELENOP was already known in the art (Sun) and further because supplementation for treatment of AITD was already known, as was the need to maintain a certain level of selenium for one's health, therefore one would expect success applying the determination to monitor levels (Ventura).

Regarding claim 4, Sun is teaching step a2 (SELENOP) measured using immunoassay (ELISA, see cited above).

Regarding claim 5, Sun teach the assay methods including Xray fluorescence spectroscopy (see cited above).

Regarding claim 6, the combination of Sun in view of Ventura is teaching evaluating one's risk of Se deficiency subsequent to the measurement in relation to ATDD (i.e., as a companion diagnostic method to monitor a need for Se supplementation). Specifically, to ensure one's supplementation is needed and not exceeding amounts known to be advantageous (as discussed above, the combination of the cited art appreciates that too high can have adverse effects). As such, the combination of the cited art addresses using the method in the quantification of Se requirement and dosage during supplementation, thereby also addressing claim 7.

9. Claim(s) 3, 9, 10 and 20 are rejected under 35 U.S.C. 103 as being unpatentable over Sun et al. and Ventura et al., as applied to claim 1 above, and further in view of Jimenez, Alexander. "Autoimmune Thyroid Diseases Associated with Infections", Nov. 29, 2017. https://www.linkedin.com/pulse/autoimmune-thyroid-diseases-associated-infections-jimenez-. Accessed: 06/23/2026.

Sun et al. and the cited art teach a method substantially as claimed, however, Sun et al. does not teach the subject is a subject suffering from a disorder selected from the list recited at claim 3 (the list including "infection").

See Jimenez, according to numerous research studies, autoimmune thyroid diseases can be caused due to a variety of factors (page 1), Jimenez teaching one autoimmune thyroid disease trigger is infection (page 2, paragraph 2). Jimenez refers to three theories linking infections and autoimmune thyroid diseases (pages 2-3). Some examples of infections linked with autoimmune thyroid diseases are presented in Jimenez (see pages 3-4, viral infections, bacterial infections).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified Sun et al. and the cited prior art in order to perform the assay on subjects with autoimmune thyroid disease, who are also subjects with infection, either viral or bacterial (claims 3 and 10), because it was known in the art that autoimmune thyroid disease can be triggered by infection, as such it would be expected (one having a reasonable expectation of success) that at least some subjects having AITD also be subjects with infection. Further one would have an expectation of success given the art recognized link between AITD and treating subjects with AITD with selenium supplementation.

Even further, regarding claim 9, it would have been obvious to have used the method (measuring Se, SELENOP and aAB-SELENOP, one motivated to apply supplementation as a treatment for AITD for the reasons as indicated above) in the treatment of a subject suffering from infection and having AITD in order to monitor Se levels as administered and ensure levels remain in the range as indicated by Ventura as an advantageous range (thereby addressing the broad language "using the method as defined in claim 1"). One having ordinary skill in the art would have a reasonable expectation of success given that it is known in the art that those with AITD benefit from treatment with Se supplementation, that it is known that levels should be kept in the indicated range (too little or too much considered problematic, see Ventura), and that it is known that infection is a trigger for autoimmune diseases including autoimmune thyroid diseases (Jimenez).

Regarding claim 20, see Regarding claim 4, Sun is teaching step a2 (SELENOP) measured using immunoassay (ELISA, see cited above).

10. Claim(s) 8 is rejected under 35 U.S.C. 103 as being unpatentable over Sun et al. in view of Ventura et al., as applied to claim 1 above, and further in view of CDC. "Selenium 2. Relevance to Public Health." June 25, 2018. Internet Archive Wayback Machine, https://web.archive.org/web/20180625230615/https:/www.atsdr.cdc.gov/toxprofiles/tp92-c2.pdf. Accessed: 06/23/2026.

Sun et al. and the cited prior art teach a method substantially as claimed, namely a method of evaluating a subject's health risk for Se deficiency, the combination of the cited art providing motivation to administer Se to treat AITD, performing the assay to monitor one's selenium levels to ensure proper supplementation (as the art recognizes too little and too much as having adverse effects, the prior art motivating one to supplement to the prior art indicated desired range, see as cited in detail previously above).

However, the prior art fails to teach using the method (assaying the values) in an effort to analyze and monitor Se poisoning.

CDC teach following chronic oral exposure to excessive amounts of organic selenium compounds in food (selenosis), the two principal clinical conditions observed in humans are dermal and neurological effects (see teaching the dermal manifestations of selenosis include loss of hair, deformation and loss of nails, and discoloration and excessive decay of teeth, while neurological effects include numbness, paralysis, and occasional hemiplegia, page 17, paragraph 1).

Also as noted previously above, regarding selenium supplementation, the prior art recognized, supplementation should be recommended to patients with low levels of selenium, that on the other hand, high selenium intake in individuals without proved deficit may have

important adverse effects such as hyperglycaemia and atherosclerosis (Ventura et al., page 7, col. 1, para 3).

As discussed above previously, the combination of Sun et al. and Ventura et al., there is motivation to administer supplementation and monitor biomarker levels (including that of selenium as in Sun) to determine the appropriate dose, and further it was understood that excessive exposure can lead to selenosis (CDC). It would have been further *prima facie* obvious to use the method of Sun et al. and the cited art, to analyze or monitor potential Se poisoning. Specifically, considering overconsumption of Selenium is a concern (CDC), it would have been *prima facie* obvious to use the method to monitor/analyze a subject's measured levels, for example to ensure/prevent a subject's measured Se (as measured using the assay of Sun) from exceeding the recommended level, i.e., to further apply the method for use in analyzing/monitoring Se poisoning (i.e., over administration/supplementation consumption), one further motivated to apply the method to monitor Se poisoning in order to modify dosage to return/maintain levels to the desired range for treatment without additional risk of adverse reaction from excessive amounts. One having ordinary skill would have a reasonable expectation of success applying the method for this use because it is the level of Se that is important (as per Ventura and the CDC).

11. Claim(s) 2, 13-16 are rejected under 35 U.S.C. 103 as being unpatentable over Sun et al. in view of Ventura et al., as applied to claim 1, and further in view of Turner et al.., Multianalyte Assays with Algorithmic Analysis in Women's Health, Association for Diagnostics & Laboratory Medicine, July 1, 2018, Online:

https://myadlm.org/cln/articles/2018/july/multianalyte-assays-with-algorithmic-analysis-in-womens-health, (6 pages).

Sun et al. and the cited prior art teach an assay method substantially claimed, the method evaluating the health risk of selenium deficiency in a subject (see above), however, Sun et al. and the cited prior art fail to teach the method comprising steps "setting the two or more biological parameters" (i.e., the measured biological values from claim 1 steps a and b) and calculating a risk index score for risk of Se deficiency of said subject (see rejections as detail above, under 35 U.S.C. 112(b), for the purpose of applying prior art, the claim is interpreted as taking the measured values from claim 1 and using the values as variables to calculate a risk index score representative of one's risk of Se deficiency).

Turner et al. teach for many disease no single biomarker test offers acceptable clinical sensitivity and specificity, so clinical laboratories have turned to combinations of biomarkers to better reveal truth about a patient's condition (referring to multianalyte assays with algorithmic analyses, MAAAs, tests which combine two or more markers along with patient demographics and clinical information, into an algorithm to generated diagnostic, prognostic or predictive information for a disease, i.e., a patient specific risk score, see paragraphs 1-2 of page 1, see also page 4, conclusions paragraph 1). Turner teach MAAAs can improve or refine disease detection through individual risk assessment.

It would have been *prima facie* obvious to one having ordinary skill in the prior art before the effective filing date of the claimed invention to have modified the method as taught by Sun et al. and the cited prior art, to generate a risk score (by way of MAAA, namely to have input the measured values to calculate a risk score, which reads on the claimed "risk index score"), one motivated to calculate a risk score in an effort to achieve desirable sensitivity and specificity

afforded using a multianalyte approach as compared to one single biomarker, one further motivated as this is recognized in the art as a tool to provide individual risk assessment (patient specific assessment). One having ordinary skill in the art would have a reasonable expectation of success since it is understood from Turner et al. that MAAAs as applicable to any multimarker clinical assay technique (useful for clinical diagnostics).

Regarding claim 13, Sun is teaching step a2 (SELENOP) measured using immunoassay (ELISA, see cited above).

Regarding claim 14, Sun teach the assay methods including Xray fluorescence spectroscopy (see cited above).

Regarding claim 15, the combination of Sun in view of Ventura is teaching evaluating one's risk of Se deficiency subsequent to the measurement in relation to ATDD (i.e., as a companion diagnostic method to monitor a need for Se supplementation). Specifically, to ensure one's supplementation is needed and not exceeding amounts known to be advantageous (as discussed above, the combination of the cited art appreciates that too high can have adverse effects). As such, the combination of the cited art addresses using the method in the quantification of Se requirement and dosage during supplementation, thereby also addressing claim 16.

12. Claim(s) 17 is rejected under 35 U.S.C. 103 as being unpatentable over Sun et al. in view of Ventura et al., and Turner et al., as applied to claim 2 above, and further in view of CDC. "Selenium 2. Relevance to Public Health." June 25, 2018. Internet Archive Wayback Machine, https://web.archive.org/web/20180625230615/https:/www.atsdr.cdc.gov/toxprofiles/tp92-c2.pdf. Accessed: 06/23/2026.

Sun et al. and the cited prior art teach a method substantially as claimed, namely a method of evaluating a subject's health risk for Se deficiency, the combination of the cited art providing motivation to administer Se to treat AITD, performing the assay to monitor one's selenium levels to ensure proper supplementation (as the art recognizes too little and too much as having adverse effects, the prior art motivating one to supplement to the prior art indicated desired range, see as cited in detail previously above).

However, the prior art fails to teach using the method (assaying the values) in an effort to analyze and monitor Se poisoning.

CDC teach following chronic oral exposure to excessive amounts of organic selenium compounds in food (selenosis), the two principal clinical conditions observed in humans are dermal and neurological effects (see teaching the dermal manifestations of selenosis include loss of hair, deformation and loss of nails, and discoloration and excessive decay of teeth, while neurological effects include numbness, paralysis, and occasional hemiplegia, page 17, paragraph 1).

Also as noted previously above, regarding selenium supplementation, the prior art recognized, supplementation should be recommended to patients with low levels of selenium, that on the other hand, high selenium intake in individuals without proved deficit may have important adverse effects such as hyperglycaemia and atherosclerosis (Ventura et al., page 7, col. 1, para 3).

As discussed above previously, the combination of Sun et al. and Ventura et al., there is motivation to administer supplementation and monitor biomarker levels (including that of selenium as in Sun) to determine the appropriate dose, and further it was understood that excessive exposure can lead to selenosis (CDC). It would have been further *prima facie* obvious

to use the method of Sun et al. and the cited art, to analyze or monitor potential Se poisoning. Specifically, considering overconsumption of Selenium is a concern (CDC), it would have been obvious to use the method to monitor/analyze a subject's measured levels, for example to ensure/prevent a subject's measured Se (as measured using the assay of Sun) from exceeding the recommended level, i.e., to further apply the method for use in analyzing/monitoring Se poisoning (i.e., over administration/supplementation consumption), one further motivated to apply the method to monitor Se poisoning in order to modify dosage to return/maintain levels to the desired range for treatment without additional risk of adverse reaction from excessive amounts. One having ordinary skill would have a reasonable expectation of success applying the method for this use because it is the level of Se that is important (as per Ventura and the CDC).

13. Claim(s) 12, 18 and 19 are rejected under 35 U.S.C. 103 as being unpatentable over Sun et al. and Ventura et al., and Turner et al., as applied to claim 2 above, and further in view of Jimenez, Alexander. "Autoimmune Thyroid Diseases Associated with Infections", Nov. 29, 2017. https://www.linkedin.com/pulse/autoimmune-thyroid-diseases-associated-infections-jimenez-. Accessed: 06/23/2026.

Sun et al. and the cited art teach a method substantially as claimed, however, Sun et al. does not teach the subject is a subject suffering from a disorder selected from the list recited at claim 3 (the list including "infection").

See Jimenez, according to numerous research studies, autoimmune thyroid diseases can be caused due to a variety of factors (page 1), Jimenez teaching one autoimmune thyroid disease trigger is infection (page 2, paragraph 2). Jimenez refers to three theories linking infections and

autoimmune thyroid diseases (pages 2-3). Some examples of infections linked with autoimmune thyroid diseases are presented in Jimenez (see pages 3-4, viral infections, bacterial infections).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified Sun et al. and the cited prior art in order to perform the assay on subjects with autoimmune thyroid disease, who are also subjects with infection, either viral or bacterial (claims 12 and 19), because it was known in the art that autoimmune thyroid disease can be triggered by infection, as such it would be expected (one having a reasonable expectation of success) that at least some subjects having AITD also be subjects with infection. Further one would have an expectation of success given the art recognized link between AITD and treating subjects with AITD with selenium supplementation.

Even further, regarding claim 18, it would have been obvious to have used the method (measuring Se, SELENOP and aAB-SELENOP, one motivated to apply supplementation as a treatment for AITD for the reasons as indicated above) in the treatment of a subject suffering from infection and having AITD in order to monitor Se levels as administered and ensure levels remain in the range as indicated by Ventura as an advantageous range (thereby addressing the broad language "using the method as defined in claim 1"). One having ordinary skill in the art would have a reasonable expectation of success given that it is known in the art that those with AITD benefit from treatment with Se supplementation, that it is known that levels should be kept in the indicated range (too little or too much considered problematic, see Ventura), and that it is known that infection is a trigger for autoimmune diseases including autoimmune thyroid diseases (Jimenez).

14. Claim 11 is rejected under 35 U.S.C. 103 as being unpatentable over Mita et al., Selenoprotein P-neutralizing antibodies improve insulin secretion and glucose sensitivity in type 2 diabetes mouse models, Nature communications, 8(1658), (2017), 8 pages in view of Zuk et al., US 4,208,479 and Pruess et al., WO2010/085345 A1.

Regarding claim 11, Mita et al. teach ninety-six well microtiter plates coated with hEP protein (hSEP, purified human SEP, i.e., selenoprotein P) before binding and detecting anti-hSEP antibody (see page 14, col. 2, para 2). The assay described by Mita (the direct ELISA), is considered usable to perform the method of claim 1, in that it is usable to detect selenop-Ab (antibodies against SELENOP) of claim 1. The present claim describes the kit structurally as comprising specific reagent for determination of antibodies against SELENOP and a user leaflet, these two components recited as the only structural components required, as the remaining limitations are recited as optional (i.e., see c, "optionally one or more reagents or chemicals selected from buffer, calibration standard, negative control, positive control" and see d, "optionally a specific reagent for determination of SELENOP).

Mita, while does teach specific reagent for the determination of antibodies against SELENOP (i.e., hSEP coated wells), fails to recite "user leaflet" and fails to recite the assay components provided in the form of a "kit".

Regarding the limitation "kit", Zuk et al. teach that in performing assays, it is convenient and to combine the necessary reagents together in a kit (column 22, lines 20-68 in particular). Zuk et al. further teach that this may improve assay accuracy.

Therefore, it would have been obvious to one of ordinary skill in the art to have provided the reagent as taught by Mita (reagent that is hsep coated wells for detection of anti-hSEP) as a kit form for convenience and accuracy as taught by Zuk et al.

Further, regarding "leaflet", see MPEP 2111.05, I., B. Where a product merely serves as a support for printed matter, no functional relationship exists. Additionally, where the printed matter and product do not depend upon each other, no functional relationship exists. For example, in a kit containing a set of chemicals and a printed set of instructions for using the chemicals, the instructions are not related to that particular set of chemicals. *In re Ngai*, 367 F.3d at 1339, 70 USPQ2d at 1864.

In the present case, "leaflet" is interpreted as meaning "printed matter", and in the present case, the printed matter (leaflet) and the product do not depend on each other, as no functional relationship exists. The leaflet is not related to the particular reagent that is the specific reagent for determination of antibodies against SELENOP.

Nonetheless, see Preuss teach providing diagnostic reagent kits, teaching kit including instructions or other printed material on how to use the various components of the kits for diagnostic purposes (page 31, lines 29-30).

Specifically, as shown by Preuss, it was known to include instruction, it would have been further prima facie obvious to have include instructions (i.e., a leaflet) in order to instruct a user how to use the components of the kit to achieve detection. One having a reasonable expectation of success because including instructions for performing an assay with the reagents was a known technique at the time (as is supported by Pruess).

Correspondence

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ELLEN J MARCSISIN whose telephone number is (571)272-6001. The examiner can normally be reached M-F 8:00am-4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached at 571-272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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